

Introduction to Tensor Analysis and CMS PS Ch 11

Jiwoo Kwak

May 5, 2026

1 Properties of the Surface Covariant Derivative

1.1 Exercise229.

Show the tensor property of the surface covariant derivative.

sol.

Let's start by recalling the definition of the surface covariant derivative.

$$\nabla_\nu T_\beta^\alpha = \frac{\partial T_\beta^\alpha}{\partial S^\nu} + \Gamma_{\nu\omega}^\alpha T_\beta^\omega - \Gamma_{\nu\beta}^\omega T_\omega^\alpha. \quad (1)$$

To show the tensor property of the given expression, we need to change all the indices with prim indices and then revert to the original indices with Jacobian matrices. The transformation law may then be expressed in terms of partial derivatives of the coordinate functions, which means

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = \left(\frac{\partial T_\beta^\alpha}{\partial S^\nu} + \Gamma_{\nu\omega}^\alpha T_\beta^\omega - \Gamma_{\nu\beta}^\omega T_\omega^\alpha \right) J_\alpha^{\alpha'} J_{\beta'}^\beta J_{\nu'}^\nu.$$

or more simply

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = (\nabla_\nu T_\beta^\alpha) J_\alpha^{\alpha'} J_{\beta'}^\beta J_{\nu'}^\nu. \quad (2)$$

By putting prime indices on 1, we get

$$\nabla_{\nu'} T_{\beta'}^{\alpha'} = \frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} + \Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} - \Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'}. \quad (3)$$

And now look at the terms each by each.

$$\frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} \quad (4a)$$

$$\Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} \quad (4b)$$

$$\Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'} \quad (4c)$$

(4a)

$$\begin{aligned} \frac{\partial T_{\beta'}^{\alpha'}}{\partial S^{\nu'}} &= \frac{\partial (T_\beta^\alpha J_\alpha^{\alpha'} J_{\beta'}^\beta)}{\partial S^{\nu'}} = \frac{\partial T_\beta^\alpha}{\partial S^{\nu'}} J_\alpha^{\alpha'} J_{\beta'}^\beta + T_\beta^\alpha \frac{\partial J_\alpha^{\alpha'}}{\partial S^{\nu'}} J_{\beta'}^\beta + T_\beta^\alpha J_\alpha^{\alpha'} \frac{\partial J_{\beta'}^\beta}{\partial S^{\nu'}} \\ &= \frac{\partial T_\beta^\alpha}{\partial S^\nu} (J_\alpha^{\alpha'} J_{\beta'}^\beta J_{\nu'}^\nu) + T_\beta^\alpha J_{\alpha\omega}^{\alpha'} J_{\beta'}^\beta J_{\nu'}^\omega + T_\beta^\alpha J_\alpha^{\alpha'} J_{\beta'\nu'}^\beta. \end{aligned} \quad (5)$$

Before treating 4b and 4c, let us look at the the definition of the **Christoffel symbols on embedded surfaces**.

$$\Gamma_{\beta\nu}^{\alpha} = \mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\beta}}{\partial S^{\nu}} \quad (6)$$

(4b)

$$\begin{aligned} \Gamma_{\nu'\omega'}^{\alpha'} T_{\beta'}^{\omega'} &= \mathbf{S}^{\alpha'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\omega'}} T_{\beta'}^{\omega'} = \mathbf{S}^{\alpha'} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\omega'}} T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\beta'}^{\beta} \\ &= \left(\mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\omega'}} J_{\nu'}^{\nu} + \mathbf{S}^{\alpha} \cdot \mathbf{S}_{\nu} \frac{\partial J_{\nu'}^{\nu}}{\partial S^{\omega'}} \right) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= \left(\mathbf{S}^{\alpha} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\rho}} J_{\omega'}^{\rho} J_{\nu'}^{\nu} + \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu} \right) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= (\Gamma_{\nu\rho}^{\alpha} J_{\omega'}^{\rho} J_{\nu'}^{\nu} + \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu}) T_{\beta}^{\omega} J_{\omega'}^{\omega'} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\rho}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} \delta_{\nu}^{\alpha} J_{\nu'\omega'}^{\nu} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\rho}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} \delta_{\omega}^{\rho} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} J_{\nu'\omega}^{\alpha} \\ &= T_{\beta}^{\omega} \Gamma_{\nu\omega}^{\alpha} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega'} \\ &= \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} \left(J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\beta}^{\omega} J_{\alpha'}^{\alpha'} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega'}. \end{aligned}$$

(4c)

$$\begin{aligned} \Gamma_{\nu'\beta'}^{\omega'} T_{\omega'}^{\alpha'} &= \mathbf{S}^{\omega'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\beta'}} T_{\omega'}^{\alpha'} \\ &= \mathbf{S}^{\omega} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\beta'}} T_{\omega}^{\alpha} J_{\alpha'}^{\alpha'} J_{\omega'}^{\omega'} J_{\rho}^{\rho} \\ &= \mathbf{S}^{\omega} \cdot \frac{\partial (\mathbf{S}_{\nu} J_{\nu'}^{\nu})}{\partial S^{\beta'}} T_{\omega}^{\alpha} J_{\alpha'}^{\alpha'} \delta_{\rho}^{\omega} \\ &= \left(\mathbf{S}^{\rho} \cdot \frac{\partial \mathbf{S}_{\nu}}{\partial S^{\beta}} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + J_{\nu'\beta'}^{\rho} \right) T_{\rho}^{\alpha} J_{\alpha'}^{\alpha'} \\ &= \left(\Gamma_{\nu\beta}^{\rho} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + J_{\nu'\beta'}^{\rho} \right) T_{\rho}^{\alpha} J_{\alpha'}^{\alpha'} \\ &= T_{\rho}^{\alpha} \Gamma_{\nu\beta}^{\rho} J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha'} \\ &= \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \left(J_{\alpha'}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha'}. \end{aligned}$$

Now putting all the terms in 3, we get

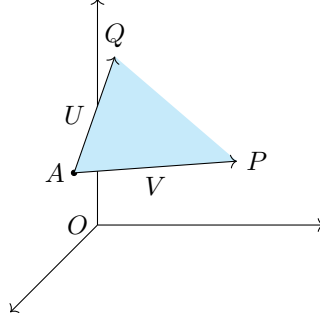
$$\begin{aligned}
\nabla_{\nu'} T_{\beta'}^{\alpha'} &= \mathbf{S}^{\omega'} \cdot \frac{\partial \mathbf{S}_{\nu'}}{\partial S^{\beta'}} T_{\omega'}^{\alpha'} \\
&= \frac{\partial T_{\beta}^{\alpha}}{\partial S^{\nu}} J_{\alpha'}^{\alpha} J_{\beta'}^{\beta} J_{\nu'}^{\nu} + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} \left(J_{\alpha'}^{\alpha} J_{\beta'\nu'}^{\beta} \right) \\
&\quad + \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} \left(J_{\alpha'}^{\alpha} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) + T_{\beta}^{\omega} J_{\alpha'}^{\alpha} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \left(J_{\alpha'}^{\alpha} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha} \\
&= \left(\frac{\partial T_{\beta}^{\alpha}}{\partial S^{\nu}} + \Gamma_{\nu\omega}^{\alpha} T_{\beta}^{\omega} - \Gamma_{\nu\beta}^{\rho} T_{\rho}^{\alpha} \right) \left(J_{\alpha'}^{\alpha} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) \\
&\quad + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} J_{\alpha'}^{\alpha} J_{\beta'\nu'}^{\beta} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha} \\
&= (\nabla_{\nu} T_{\beta}^{\alpha}) \left(J_{\alpha'}^{\alpha} J_{\beta'}^{\beta} J_{\nu'}^{\nu} \right) \\
&\quad + T_{\beta}^{\alpha} J_{\alpha\omega}^{\alpha'} J_{\beta'}^{\beta} J_{\nu'}^{\omega} + T_{\beta}^{\alpha} J_{\alpha'}^{\alpha} J_{\beta'\nu'}^{\beta} + T_{\beta}^{\omega} J_{\alpha'}^{\alpha} J_{\nu'\omega'}^{\alpha} J_{\beta'}^{\beta} J_{\omega'}^{\omega} - T_{\rho}^{\alpha} J_{\nu'\beta'}^{\rho} J_{\alpha'}^{\alpha}.
\end{aligned}$$

And since $T_{\beta}^{\alpha} J_{\alpha'}^{\alpha} J_{\beta'\nu'}^{\beta}$

2 Rules of the Game

Before we begin, consider some geometrical logic that can be applied to n-dimensional space. For the Pythagorean Theorem and law of cosines, we can treat them also in n-dimensional Euclidean space without proof; they are related to only two vectors.

In Linear Algebra, treating two vectors means that there is a subspace spanned by the vectors, and the space itself has a degree of freedom, 2 (already proved from two-dimensional Euclidean Space.) That is, no matter what basis vectors are set and how many vectors are in the space, we only cover two-dimensional space such that we can still use the properties:



Law of Cosines

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 - 2\|U\|\|V\|\cos\alpha \quad (7)$$

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 - 2U \cdot V \quad (8)$$

Pythagorean Theorem, in case of $\alpha = 0^\circ$

$$\|U - V\|^2 = \|U\|^2 + \|V\|^2 \quad (9)$$

From the Law of Cosines, we can also find the formula for the dot product without the angle:

$$U \cdot V = \frac{\|U\|^2 + \|V\|^2 - \|U - V\|^2}{2}$$

$$U \cdot (-V) = \frac{\|U\|^2 + \|V\|^2 - \|U + V\|^2}{2}$$

which gives the formula

$$U \cdot V = \frac{\|U + V\|^2 - \|U - V\|^2}{4} \quad (10)$$

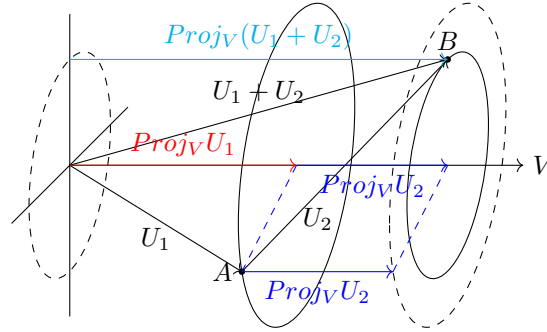
2.1 Exercise 6.

To demonstrate geometrically that the dot product is distributive, we can use the properties of vectors and the geometric interpretation of the dot product.

sol.

Consider three vectors U_1 , U_2 and V . Since the dimension of a subspace that can be made by them is at most three, they can be treated as vectors in three-dimensional space by a well-defined basis.

Now, they are represented in a cylinder parallel to the direction of V :



From the figure, Projection of $U_1 + U_2$ now has same direction with $Proj_V U_1$ and $Proj_V U_2$ such that the following holds:

$$Proj_V(U_1+U_2) = Proj_V U_1 + Proj_V U_2 \longleftrightarrow \|V\| (Proj_V(U_1+U_2)) = \|V\| (Proj_V U_1 + Proj_V U_2)$$

$$\longrightarrow (U_1 + U_2) \cdot V = U_1 \cdot V + U_2 \cdot V$$

■

References

- [1] Pavel Grinfeld. *Introduction to Tensor Analysis and the Calculus of Moving Surfaces*. Springer, 2013.
- [2] StackExchange. *Proving that the dot product is distributive?* 2015. URL: <https://math.stackexchange.com/questions/1109142/proving-that-the-dot-product-is-distributive> (visited on 07/19/2025).